

Market Microstructure Frameworks for Spoofing Detection: Isolating Manipulative Intent from Legitimate Liquidity Provision

Daniele Maria Di Nosse^{†1}, Francesco Gigante², Antonio Russo², and Fabrizio Lillo¹

¹Scuola Normale Superiore, Pisa, Italy

²Consob, Rome, Italy

July 12, 2026

[†]Corresponding author. Email: `daniele.dinosse@sns.it`

Abstract

In this paper we detect the bad guys and we find how to tell them they are actually bad guys. Hopefully, we will be able to do this without accusing the good guys of being bad guys.

Keywords. Spoofing detection; market manipulation; limit order book; high-frequency trading; anomaly detection; market maker; market microstructure

1 Introduction

The transition of global financial exchanges to fully electronic, continuous double auction matching engines has fundamentally transformed the mechanics of price discovery and liquidity provision. In modern financial ecosystems, the Limit Order Book (LOB) serves as the central nervous system of the market, aggregating the continuous stream of bid and ask limit orders submitted by diverse market participants. While the advent of algorithmic and high-frequency trading (HFT) has undeniably narrowed bid-ask spreads and enhanced short-term market efficiency, it has concurrently introduced profound vulnerabilities. The microsecond latency environment allows sophisticated algorithms to exploit the structural mechanics of the LOB, executing complex manipulative strategies that deceive heuristic-based trading algorithms and human traders alike. Among these manipulative practices, spoofing represents one of the most pervasive and challenging threats to market integrity. Spoofing is a deceptive algorithmic strategy wherein a market participant places a massive, non-bona fide order on one side of the LOB with the explicit intent of canceling it before execution. The presence of this immense, artificial volume creates a false impression of directional market pressure. This artificial pressure manipulates the micro-price and induces other market participants—particularly algorithms that rely on order book imbalance indicators—to adjust their quotes or submit market orders. Once the price has been artificially driven to a favorable level, the spoofer executes a smaller, bona fide

order on the opposite side of the book and immediately withdraws the massive deceptive orders, thereby profiting from the artificially induced price dislocation. The European Securities and Markets Authority (ESMA), through the Market Abuse Regulation (MAR) [3], strictly prohibits such practices, categorizing them as behaviors that disrupt the functioning of the trading system and create misleading signals regarding the supply and demand of financial instruments. For regulatory and supervisory authorities such as the Commissione Nazionale per le Società e la Borsa (CONSOB), the detection and prosecution of spoofing present a severe analytical challenge. The physical actions comprising a spoofing strategy—posting limit orders, updating quotes, and rapidly canceling orders—are mechanically indistinguishable from the legitimate actions of a bona fide market maker providing liquidity. Both spoofers and market makers exhibit exceedingly high order-to-trade ratios (OTTR) and operate at high frequencies. Traditional market surveillance mechanisms, which historically relied on static heuristic thresholds, volumetric alerts, or simple cancellation ratios, are fundamentally inadequate in this environment. These legacy systems generate an overwhelming volume of false positives, frequently flagging legitimate liquidity provision and inventory management as manipulative market abuse.

2 The Microstructural Divergence: Market Making Versus Spoofing

To construct a mathematical model capable of detecting market manipulation, it is first necessary to rigorously define the divergent objective functions that govern legitimate market making and spoofing. While both participants generate massive volumes of limit order messages and cancellations, their underlying economic motivations, risk profiles, and optimizations are fundamentally opposed.

2.1 The Market Maker’s Problem

A market maker functions as a continuous liquidity provider, simultaneously quoting resting limit orders at the best bid and the best ask. The primary objective function of a market maker is to maximize the utility of gains derived from capturing the bid-ask spread across a vast number of transactions, while concurrently minimizing the variance of their portfolio and strictly managing inventory risk. In market microstructure theory, the pricing dynamics of a liquidity provider are governed by the interplay between the arrival rate of market orders and the accumulation of inventory.

The optimization problem for a market maker is frequently modeled using stochastic control techniques and Hamilton-Jacobi-Bellman (HJB) equations, where the dealer sets optimal quotes to balance the asymmetric arrival of buy and sell orders. ¹ If a market maker accumulates a long position (positive inventory), they are exposed to the risk of the underlying asset depreciating. To mitigate this inventory risk, the rational market maker will lower their quotes downwards—lowering their ask price to incentivize incoming buyers, and lowering their bid price to dissuade incoming sellers. ² The bid and ask prices are dynamically adjusted functions of the mid-price, the base spread required for profitability, and an inventory penalty parameter that increases as the dealer’s position deviates from neutrality.

Consequently, a market maker’s high cancellation rate is entirely reactive and protective. Limit orders are rapidly canceled and re-posted to reflect minute changes in the fundamental mid-price, to defend against adverse selection informed traders, and to recalibrate inventory penalty. The massive volume of orders and cancellations produced by a market maker is therefore characterized by symmetry across the LOB and a continuous presence at the top of the book to execute trades. Their expected directional gain from posting an order is near zero, as their profit is entirely derived from the spread rather than from inducing a directional shift in the mid-price.

2.2 The Spoofer’s Problem

In stark contrast, a spoofing agent does not seek to capture the bid-ask spread, nor do they operate as symmetric liquidity provider. A spoofer’s objective function is purely directional. The spoofer desires to execute a relatively small, bona fide order at a highly favorable price by artificially manipulating the micro-price dynamics of the LOB. To execute this strategy, the spoofer deploys a massive non-bona fide order on the opposite side of the LOB. This deliberate injection of spurious volume creates an extreme unbalanced environment, artificially inflating the visible depth of the book. In modern electronic markets, countless algorithmic traders utilize strategies that weight the volume at the bid against the volume at the ask to predict short-term price distributions. By overwhelming one side of the book, the spoofer pollutes these volume imbalance metrics, causing HBL algorithms to mathematically deduce an impending price shift and aggressively alter their own quoting behavior.

The critical mathematical dilemma for the spoofer lies in the placement of the non-bona fide order. The spoofer must balance the expected market manipulation gain against the explicit risk of unintended execution. If the spoofer places the massive non-bona fide order exactly at the top of the book (the best bid or ask), they generate the maximum psychological and algorithmic impact on the micro-price. However, doing so exposes the spoofer to massive adverse selection and inventory risk; an incoming aggressive market order could inadvertently execute against the spoofer’s massive fake order, leaving the spoofer holding an enormous, unwanted position. To optimize this expected cost function, sophisticated spoofers rarely post their deceptive orders at the absolute best price. Instead, they optimize the posting distance (measured in ticks or basis points) deeper within the book. By placing the order at Level 2, Level 3, or beyond, the spoofer ensures the volume is still visible to the matching engine’s data feed—and thus factored into the volume imbalance metrics of victim algorithms—while maintaining a structural buffer against immediate execution. Once the intended directional shift occurs and the spoofer’s small bona fide order is executed, the massive non-bona fide order is instantly canceled. The spoofer’s cancellation intent is not to manage inventory, but to terminate the manipulative episode the exact millisecond the cost optimization concludes.

3 Existing Approaches

The literature on spoofing detection is still relatively small, but it has moved from transparent microstructure rules toward sequence and graph models that try to capture richer strategic

Table 1: Core Differentiators for Surveillance Models

Microstructure Metric	Legitimate Market Maker	Manipulative Spoofing Agent
Primary Objective Function	Maximize bid–ask spread capture while minimizing inventory variance.	Minimize execution cost of a directional trade via artificial micro-price impact.
Order Book Imbalance	Generally symmetric, adapting quotes to maintain a neutral inventory.	Extremely asymmetric, deliberately creating massive, sudden imbalances (unbalanced quoting).
Cancellation Motivation	Reactive: mitigating adverse selection and updating quotes to track fair value.	Proactive: terminating the deceptive posture immediately after a targeted execution.
Posting Distance Optimization	Top of the book (Level 1) to maximize the probability of spread execution.	Deeper in the book (Level 2/3) to maximize visibility while minimizing execution risk.
Expected Directional Gain	Zero or negative (willing to pay the spread to lay off excessive inventory risk).	Strictly positive (the massive order statistically subsidizes the execution of the small order).

behavior. For the purpose of this paper, the most useful distinction is not simply between “statistical” and “machine-learning” models. It is between methods that identify a stylized spoofing episode from order-book mechanics, methods that monitor abnormal order-flow states, and methods that learn temporal or relational patterns from proprietary event data. Figure 1 summarizes this taxonomy.

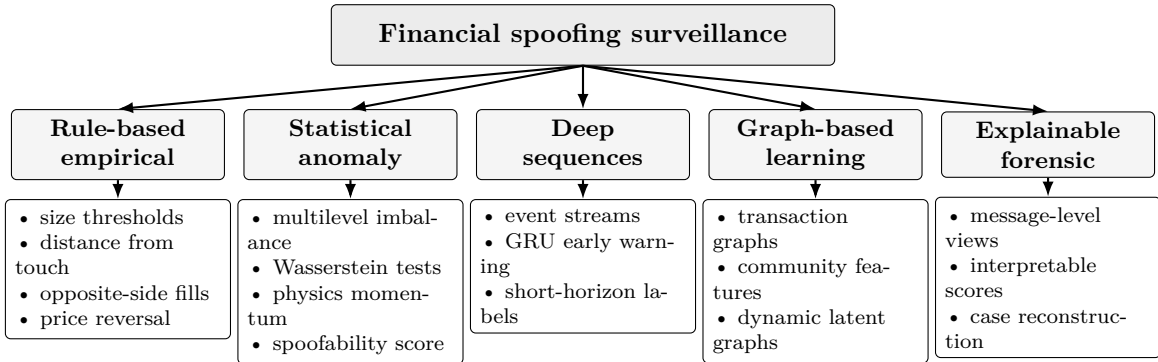


Figure 1: Method families in the spoofing-detection literature. The categories are not mutually exclusive, but they highlight the main modeling choice made by each strand of work.

Early empirical papers define spoofing through economically interpretable order-book conditions. The seminal study by [7] identifies large orders placed far from the best quote, followed by opposite-side trading and subsequent withdrawal, using account-level data from the Korea Exchange. Similar surveillance logic appears in Brazilian equities [9], FX spot markets [11],

cross-market FX settings [10], and Taiwan derivatives [1]. These studies are valuable because the alert logic can be explained to supervisors: suspicious activity is tied to order size, posting distance, cancellation timing, opposite-side execution, and short-lived price impact. Their main limitation is that thresholds are venue-specific and often depend on privileged message data or account identifiers.

A second line develops statistical and anomaly-oriented monitoring. [12] derive spoofing incentives from multilevel order-book imbalance and propose Wasserstein-distance monitoring for real-time detection. [8] use a statistical-physics representation of the order book to detect abnormal momentum in crypto markets. More recent explainable approaches also fall partly in this family: forensic visualization reconstructs message-level spoofing episodes in prosecuted cases [2], while interpretable probabilistic neural networks estimate the conditional gain from manipulating a centralized crypto order book [4]. These methods improve on pure threshold rules by modeling the state of the book, but many papers still report limited benchmark metrics or rely on non-public datasets.

Machine-learning work mainly adds temporal or relational structure. [13] formulate spoofing as an early-warning problem and train a GRU model on cryptocurrency order-book event streams. Graph-based methods then move the unit of analysis from isolated orders to networks of traders and transactions, which is important for detecting coordinated or “conspiracy” spoofing [5, 6, 14]. The strongest reported graph models combine temporal transaction graphs, community-level features, and dynamic latent representations, but their practical value is difficult to assess without open labels and chronological cross-market validation.

Overall, the literature suggests two requirements for a robust surveillance model. First, the features should remain microstructure-grounded: a detector should know whether suspicious volume is large, distant from the touch, rapidly cancelled, and conditional on an opposite-side execution. Second, the evaluation should respect the enforcement setting: labels are noisy, true positives are rare, and false positives impose substantial analyst costs. This motivates the agent-level, spatio-temporal framework developed below, which keeps the interpretability of rule-based surveillance while using the full order-flow sequence available in proprietary data.

4 Data

The empirical diagnostics reported below are computed on three CONSOB limit-order-book message samples: Risanamento (June–November 2024), Nexi (23 April 2024), and Ferrari (13 June 2024). For each instrument we reconstruct the visible order book from order-level events, infer the tick size from the reconstructed best quotes, and compute the top-5 episode-conditioned metrics using a 10-second post-execution cancellation window and a 90-second maximum age for candidate opposite-side orders. The resulting objects are surveillance alerts for analyst review rather than enforcement labels.

5 Model

In this section we provide the base reasoning and metrics used to distinguish between legitimate market making behaviour and possible spoofing strategies.

5.1 A Spatio-Temporal Framework for Spoofing Detection

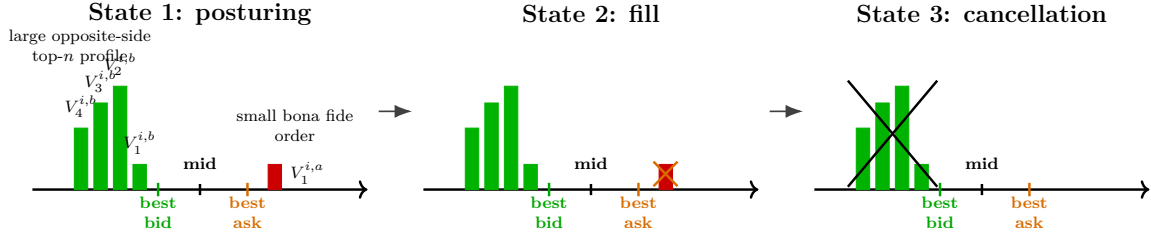


Figure 2: Schematic top- n spoofing timeline. The large non-bona fide order is represented as an agent-specific depth profile over several bid levels, while the bona fide order sits close to the opposite best quote. The same construction applies symmetrically to sell- and buy-intent episodes.

To rigorously distinguish manipulative spoofing from legitimate liquidity provision, we define the microstructural signature of the strategy as a conditional deformation of the agent’s limit-order-book footprint. The relevant object is not only the size of a single order at the best bid or ask, but the whole agent-specific liquidity profile over the first n price levels. This is important because a sophisticated spoofer may place the non-bona fide order away from the absolute best quote, for example at Level 2, Level 3, or across several adjacent levels, in order to remain visible while reducing the probability of unintended execution.

We formalize the execution timeline of a spoofing agent i into three states:

1. **State 1 (Deceptive Posturing):** shortly before the execution, the agent posts a large non-bona fide liquidity profile on one side of the book, typically away from the touch, while simultaneously maintaining a smaller bona fide order close to the opposite best quote.
2. **State 2 (Execution):** the artificial depth profile distorts short-horizon imbalance signals and the bona fide order is executed.
3. **State 3 (Risk Termination):** immediately after the execution, the agent cancels the opposite-side liquidity profile, thereby terminating the execution risk associated with the deceptive orders.

The model below generalizes the best-level construction to the top n levels while preserving the same economic logic: a large asymmetric profile must be posted shortly before the fill, must be present before the fill, and must disappear conditionally after the fill. This timing restriction is important: long-lived resting liquidity that happens to be on the opposite side of a later execution should not be mechanically interpreted as part of a spoofing episode.

5.1.1 Top- n Agent-Specific Depth Profiles

Let $s \in \{b, a\}$ denote the bid or ask side of the book, and let $k = 1, \dots, n$ denote the book level, with $k = 1$ corresponding to the same-side best quote. For agent i , define

$$V_k^{i,s}(t) \tag{1}$$

as the volume posted by the agent at time t on side s and level k . We denote by

$$\mathbf{V}_n^{i,s}(t) = (V_1^{i,s}(t), \dots, V_n^{i,s}(t)) \quad (2)$$

the corresponding top- n agent-level depth vector. In applications where the book depth varies substantially across assets or intraday periods, the raw volume can be normalized by the total visible market depth at the same side and level:

$$\tilde{V}_k^{i,s}(t) = \frac{V_k^{i,s}(t)}{D_k^s(t)}, \quad (3)$$

where $D_k^s(t)$ is the total displayed market depth on side s and level k . The distance of a level from the touch is denoted by $d_k^s(t)$. In a tick-time perspective, $d_k^s(t)$ can be measured as the number of ticks between level k and the same-side best quote, shifted by one unit so that the top level has positive distance. This avoids assigning zero weight to Level 1 while preserving the ranking of deeper levels.

5.1.2 Multilevel Distance-Weighted Imbalance

The first ingredient is a scalar projection of the top- n depth profile. The kernel should assign high weight to levels that satisfy two requirements simultaneously: they remain visible to traders and algorithms that process multilevel book information, yet they are sufficiently far from the touch to carry relatively low immediate execution risk. A convenient parametric candidate that expresses this trade-off is

$$\omega_k^s(t) = \frac{e^{-\lambda d_k^s(t)} (1 - e^{-\kappa d_k^s(t)})}{\sum_{\ell=1}^n e^{-\lambda d_\ell^s(t)} (1 - e^{-\kappa d_\ell^s(t)})}, \quad (4)$$

Here $e^{-\lambda d}$ is a stylized visibility component that decreases with depth, whereas $1 - e^{-\kappa d}$ is a stylized protection component that increases as immediate execution becomes less likely. Their product balances the near-touch penalty generated by execution risk against the loss of visibility farther inside the book. The parameters κ and λ control how quickly these two components change with distance. This form is useful for stating the economic mechanism, but it imposes exponential shapes and requires the researcher to select both rates. We do not impose those restrictions in the empirical analysis; instead, we estimate the two components directly from each instrument's order flow.

Empirical depth kernel used in the analysis. Let m index the instrument and let $\mathcal{T}_m$ denote its calibration sample. For each side s and level k , we first estimate the probability that aggressive order flow reaches the price displayed at that level within a horizon H , chosen consistently with the post-execution cancellation window. Let $R_{t,k}^{m,s}(H)$ equal one if, during $[t, t + H]$, aggressive executions occur at or through the price of level k observed at time t , and zero otherwise. Thus an ask level is reached by buy-initiated aggressive flow, while a bid level is reached symmetrically by sell-initiated aggressive flow. Conditional on level k being observable,

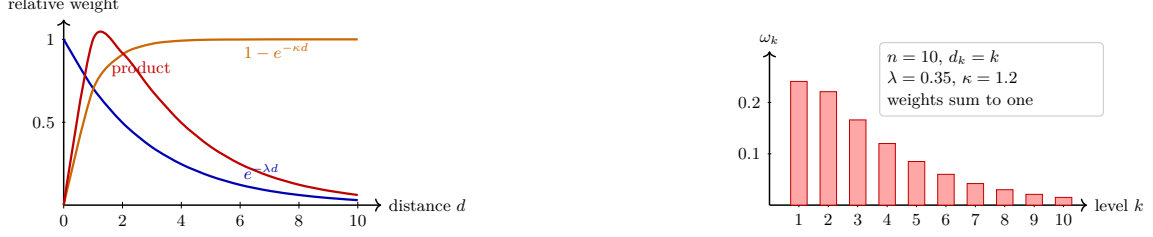


Figure 3: Conceptual illustration of the parametric candidate in [equation \(4\)](#). Visibility decreases with distance from the touch, protection from immediate execution increases, and their product represents the trade-off between the two. The right panel shows the resulting normalized weights for one illustrative parameter choice. This figure motivates the economic construction; the empirical analysis uses the instrument-specific kernel in [equation \(10\)](#) rather than these parametric weights.

its empirical hit probability is

$$\hat{p}_k^{m,s}(H) = \frac{\sum_{t \in \mathcal{T}_m} R_{t,k}^{m,s}(H)}{\sum_{t \in \mathcal{T}_m} \mathbb{I}\{\text{level } k \text{ is observable on side } s\}}. \quad (5)$$

The denominator counts the snapshots in which that level exists, while the numerator counts how often its displayed price is subsequently reached. Hence $\hat{p}_k^{m,s}(H)$ is a short-horizon execution-risk proxy. It measures level reach rather than the fill probability of an individual order, which would additionally depend on queue position and intervening cancellations. Its complement

$$\hat{\rho}_k^{m,s}(H) = 1 - \hat{p}_k^{m,s}(H), \quad (6)$$

is the empirical low-execution-risk, or protection, component: a value close to one identifies a level that is rarely reached over horizon H . Optional clipping or flooring is used only to prevent degenerate zero weights in finite samples.

The second component is an empirical proxy for visibility: whether fluctuations in the depth displayed at a level are associated with subsequent short-horizon price changes. Let $D_k^{m,s}(t)$ be total market depth, rather than agent- i depth, at level k , and define total displayed top- n depth at snapshot t as

$$D_n^m(t) = \sum_{r=1}^n \left(D_r^{m,a}(t) + D_r^{m,b}(t) \right). \quad (7)$$

The signed, normalized contribution of side s and level k to the book is

$$Z_{t,k}^{m,s} = \sigma_s \frac{D_k^{m,s}(t)}{D_n^m(t)}, \quad \sigma_s = \begin{cases} +1, & s = ask, \\ -1, & s = bid. \end{cases} \quad (8)$$

Thus $|Z_{t,k}^{m,s}|$ is the fraction of visible top- n market depth supplied by that side, while the sign records that ask depth and bid depth contribute in opposite directions to book imbalance. Let $\Delta M_{t,H}^m = M^m(t+H) - M^m(t)$ denote the subsequent mid-price change. We define level- k visibility as the magnitude of its sample covariance with that price change:

$$\hat{\nu}_k^{m,s}(H) = \left| \widehat{\text{Cov}}_{t \in \mathcal{T}_m} \left(Z_{t,k}^{m,s}, \Delta M_{t,H}^m \right) \right|. \quad (9)$$

The absolute value makes the component nonnegative: $\hat{\nu}_k^{m,s}(H)$ measures the magnitude, not the direction, of level-specific co-movement. It is a covariance-based weighting proxy, not a causal measure of price impact. Because the covariance is computed in instrument-specific price units, its magnitude is not compared across instruments; it is used only to allocate weights across levels on the same side of the same instrument.

Mirroring the economic decomposition in [equation \(4\)](#), the operational empirical kernel multiplies visibility by low-execution-risk protection and normalizes the product within each instrument and side:

$$\hat{\omega}_k^{m,s}(H) = \frac{\hat{\nu}_k^{m,s}(H) \hat{\rho}_k^{m,s}(H)}{\sum_{\ell=1}^n \hat{\nu}_\ell^{m,s}(H) \hat{\rho}_\ell^{m,s}(H)}. \quad (10)$$

Levels therefore receive high weight only when they are both informative about short-horizon book dynamics and relatively unlikely to execute immediately. In the formulas below, $\omega_k^s(t)$ is replaced by the corresponding $\hat{\omega}_k^{m,s}(H)$. All events for instrument m use the same fitted kernel; it is not re-estimated by client, event, or alert outcome. For the descriptive full-sample analysis reported here, $\mathcal{T}_m$ is the full available sample. In a prospective or out-of-sample surveillance exercise, the kernel must instead be estimated using only an earlier training period and then held fixed, so that future order flow does not enter the weights.

When a scalar summary is useful, we report the exposure-weighted log-linear decay slope of $\hat{p}_k^{m,s}(H)$ across depth as an implied $\hat{\kappa}^m$ summary of execution risk, and the observation-weighted log-linear decay slope of $\hat{\nu}_k^{m,s}(H)$ as an implied $\hat{\lambda}^m$ summary of visibility. The operational kernel, however, uses the empirical hit-probability and covariance profiles themselves rather than substituting a common exponential shape. The implied slopes summarize how closely each empirical component resembles the motivating parametric candidate; they do not determine the operational weights. This specification allows Risanamento, Nexi, and Ferrari to have different depth profiles rather than forcing a common exponential shape across instruments.

Table 2: Full-sample empirical top-5 depth-kernel calibration at horizon $H = 10$ seconds. The weights are the operational object used by the empirical runs; $\hat{\kappa}$ and $\hat{\lambda}$ are collapsed instrument-level log-decay summaries. A value of $\hat{\lambda} = 0$ means that the empirical visibility-covariance profile does not exhibit a positive exponential decay over the observed levels, not that the visibility component is zero.

Instrument	Kernel rows	$\hat{\kappa}$	$\hat{\lambda}$
Risanamento	1,004	0.823	0.000
Nexi	357	0.450	0.000
Ferrari	226	0.984	0.154

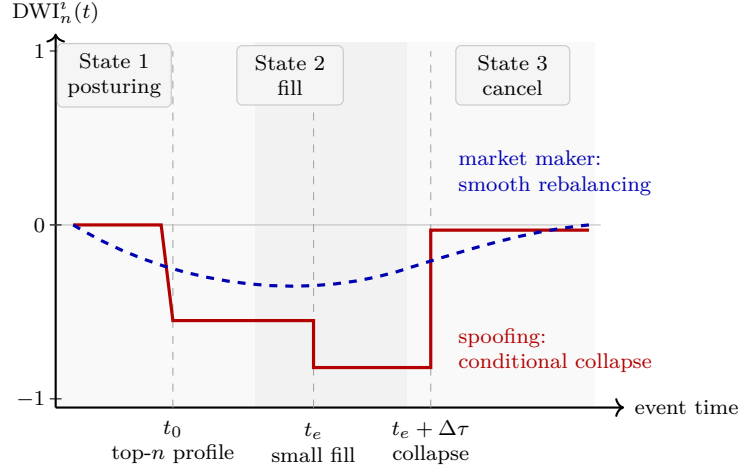
The agent's weighted top- n liquidity on side s is then

$$L_n^{i,s}(t) = \sum_{k=1}^n \omega_k^s(t) \tilde{V}_k^{i,s}(t). \quad (11)$$

We define the *Multilevel Distance-Weighted Imbalance* for agent i as

$$\text{DWI}_n^i(t) = \frac{L_n^{i,a}(t) - L_n^{i,b}(t)}{L_n^{i,a}(t) + L_n^{i,b}(t)}. \quad (12)$$

With this sign convention, $DWI_n^i(t) > 0$ indicates an ask-heavy agent profile, while $DWI_n^i(t) < 0$ indicates a bid-heavy agent profile. Because the denominator normalizes by the total weighted agent footprint, the statistic is comparable across agents and trading periods. The case $n = 1$ recovers a best-level version of the measure, whereas larger n values allow the detector to capture deceptive liquidity placed behind the touch.



For the opposite spoofing direction, the sign pattern is mirrored.

Figure 4: Stylized event-time behavior of the multilevel distance-weighted imbalance. A spoofing episode produces a large top- n imbalance and, after the small opposite-side execution, a rapid return toward zero when the deceptive profile is cancelled. Inventory rebalancing by a market maker should evolve more smoothly.

5.1.3 Conditionality and Intent Isolation

A static multilevel imbalance is not sufficient to isolate manipulative intent. A legitimate market maker may also exhibit temporary imbalance while managing inventory or reacting to adverse selection. The discriminating information is the conditionality of the subsequent cancellation: for a spoofer, the disappearance of the large liquidity profile is mechanically tied to the execution of the smaller bona fide order on the opposite side.

Let e denote an execution of agent i at timestamp t_e , and let $s_e \in \{b, a\}$ be the side of the executed bona fide order. The suspected deceptive side is the opposite side, denoted by $\bar{s}_e$. We define

$$t_e^- = t_e - \epsilon_t, \quad t_e^+ = t_e + \Delta\tau, \quad (13)$$

where ϵ_t is chosen so that t_e^- represents the book state immediately before the execution, while $\Delta\tau$ bounds the mechanical latency of the cancellation response. We also impose a pre-execution episode window $\Theta > 0$. An opposite-side order is considered part of the candidate deceptive profile for execution e only if its first observed posting time τ_o satisfies

$$0 \leq t_e - \tau_o \leq \Theta. \quad (14)$$

In the empirical implementation below we use $\Theta = 90$ seconds as an exploratory default. The candidate deceptive orders used for episode attribution are therefore restricted to same-client

top- n orders posted within this pre-execution window. This restriction is essential because a sophisticated spoofer may maintain a nearly balanced background market-making profile while adding a short-lived deceptive layer before the fill. If the conditionality score were computed on the unrestricted standing profile, the balanced background liquidity could dilute the imbalance change generated by the deceptive layer. We therefore distinguish the unrestricted top- n state, which remains useful for describing the agent's ordinary book footprint, from an *episode-conditioned* profile used for spoofing attribution.

Let $\mathcal{A}_{e,k,\Theta}^{i,s}(t)$ be the set of active orders of client i on side s and level $k \leq n$ at time t whose first observed posting time lies in $[t_e - \Theta, t_e]$. The episode-conditioned weighted liquidity is defined as

$$L_{e,n,\Theta}^{i,s}(t) = \sum_{k=1}^n \omega_k^s(t) \sum_{o \in \mathcal{A}_{e,k,\Theta}^{i,s}(t)} \tilde{q}_o(t), \quad (15)$$

where $\tilde{q}_o(t)$ denotes the visible quantity of order o , optionally normalized by the corresponding market depth as in Equation (3). The episode-conditioned imbalance is then

$$\text{DWI}_{e,n,\Theta}^i(t) = \frac{L_{e,n,\Theta}^{i,a}(t) - L_{e,n,\Theta}^{i,b}(t)}{L_{e,n,\Theta}^{i,a}(t) + L_{e,n,\Theta}^{i,b}(t) + \varepsilon}. \quad (16)$$

This object measures the incremental liquidity layer that appeared shortly before execution e , rather than the client's entire standing book. The subsequent conditionality indices are computed on this episode-conditioned profile.

We define the *Spoofing Conditionality Index* as

$$\text{SCI}_{e,n}^i = \left| \text{DWI}_{e,n,\Theta}^i(t_e^-) - \text{DWI}_{e,n,\Theta}^i(t_e^+) \right|. \quad (17)$$

However, the top- n representation allows us to be more selective than a pure imbalance difference. We can directly measure how much of the weighted liquidity on each side disappears after the fill:

$$C_{e,n}^{i,s} = \frac{\left[L_{e,n,\Theta}^{i,s}(t_e^-) - L_{e,n,\Theta}^{i,s}(t_e^+) \right]_+}{L_{e,n,\Theta}^{i,s}(t_e^-) + \varepsilon}, \quad s \in \{b, a\}, \quad (18)$$

where $[x]_+ = \max\{x, 0\}$. A spoofing episode should produce a large value of $C_{e,n}^{i,\bar{s}_e}$, because the liquidity that vanishes is expected to be on the side opposite to the bona fide execution. By contrast, a general market-maker quote refresh induced by a mid-price change should tend to affect both sides more symmetrically.

We therefore define the *Multilevel Spoofing Conditionality Index* as

$$\text{MSCI}_{e,n}^i = \underbrace{\text{SCI}_{e,n}^i}_{\text{shape change}} \underbrace{C_{e,n}^{i,\bar{s}_e}}_{\text{opposite side collapse}} \underbrace{\left[C_{e,n}^{i,\bar{s}_e} - C_{e,n}^{i,s_e} \right]_+}_{\text{asymmetry}} \quad (19)$$

This score is high only when three conditions hold simultaneously: the recently posted top- n layer changes abruptly, the collapse occurs on the side opposite to the execution, and the opposite-side collapse is stronger than the same-side collapse. Computing MSCI on the episode-conditioned profile prevents old standing liquidity from masking a newly posted deceptive layer.

This construction preserves the interpretability of the original SCI while reducing false positives generated by ordinary two-sided quote updates.

The MSCI remains conservative because it is a normalized shape-collapse score: it requires a visible change in the relative distance-weighted shape of the recently posted profile, not merely a large reduction in its absolute mass. This can miss episodes in which the suspicious evidence is not a large normalized imbalance change, but a large same-client opposite-side withdrawal following a small execution. To capture this second mechanism, define the set $\mathcal{C}_{e,n}^i$ of same-client, opposite-side candidate orders present before execution e and cancelled in $(t_e, t_e + \Delta\tau]$. For each cancelled order $o \in \mathcal{C}_{e,n}^i$, let q_o be its visible quantity immediately before cancellation and let $\delta_o = t_o - t_e$ be the cancellation delay. We use an exponentially decayed withdrawal quantity

$$W_{e,n}^i = \sum_{o \in \mathcal{C}_{e,n}^i} q_o \exp\left(-\frac{\delta_o}{\tau_w}\right), \quad (20)$$

where τ_w controls how rapidly post-execution cancellations are discounted. Let Q_e be the executed quantity and let $Q_{e,n}^{i,\bar{s}_e}$ be the pre-execution visible quantity of the candidate deceptive profile. The event-level *Withdrawal Multilevel Spoofing Conditionality Index* is

$$\text{WMSCI}_{e,n}^i = \log\left(1 + \frac{Q_{e,n}^{i,\bar{s}_e}}{Q_e + \varepsilon}\right) \log\left(1 + \frac{W_{e,n}^i}{Q_e + \varepsilon}\right) \frac{\sum_{o \in \mathcal{C}_{e,n}^i} q_o}{Q_{e,n}^{i,\bar{s}_e} + \varepsilon}. \quad (21)$$

The first term measures the size of the pre-execution opposite-side posture relative to the execution, the second term measures the speed-weighted withdrawal relative to the execution, and the last term measures how much of the candidate profile is actually removed. Thus WMSCI complements MSCI as a mass-withdrawal score: MSCI detects abrupt collapse in the normalized episode-conditioned DWI shape, while WMSCI detects small executions followed by large same-client opposite-side cancellations even when the normalized imbalance movement of the recent layer is muted.

Both indices are event-level, client-conditioned quantities. The superscript i means that the book footprint is reconstructed for client i , but $\text{MSCI}_{e,n}^i$ and $\text{WMSCI}_{e,n}^i$ are not yet client-session scores: they describe what happened around one execution e . Intuitively, MSCI asks whether the client's recently posted top- n layer changed shape in a spoofing-like way after the execution. WMSCI asks a narrower order-level mass question: after a small fill, did the same client quickly remove a large amount of the opposite-side candidate profile?

For example, suppose client i executes a buy order of $Q_e = 10$ shares while displaying 1,000 shares on the ask side within the top n levels. If the ask-heavy profile disappears and the bid-side profile is largely unchanged, then $C_{e,n}^{i,\bar{s}_e}$ is large, $C_{e,n}^{i,s_e}$ is small, and MSCI is high because the distance-weighted profile itself collapses in the suspicious direction. If instead the aggregate DWI changes little, perhaps because the cancelled orders are at levels that have limited kernel weight or because recent same-client liquidity on the other side offsets the imbalance, MSCI may be close to zero. WMSCI can still be high if, after the 10-share buy, the same client cancels 900 of the 1,000 opposite-side shares quickly. The evidentiary object is then not a large normalized imbalance jump, but the size and speed of the matched withdrawal relative to the small execution.

5.1.4 Multilevel Conditional Probability Score

To aggregate repeated behavior over a session, let N_i be the number of relevant small-order executions by agent i . For a fixed depth n , we define the *Multilevel Conditional Probability Score* as

$$\text{MCPS}_n^i = \frac{1}{N_i} \sum_{e=1}^{N_i} \mathbb{I}(\text{MSCI}_{e,n}^i > \gamma_n), \quad (22)$$

where γ_n is an empirically calibrated threshold for the selected book depth and episode window. This is the direct multilevel analogue of the original CPS: it counts the fraction of executions after which the agent exhibits a sufficiently large conditional, opposite-side collapse of its recently posted top- n liquidity layer. An analogous session statistic can be computed by replacing $\text{MSCI}_{e,n}^i$ with $\text{WMSCI}_{e,n}^i$ in Equation (22); in the implementation we report both event scores because they flag different microstructural mechanisms.

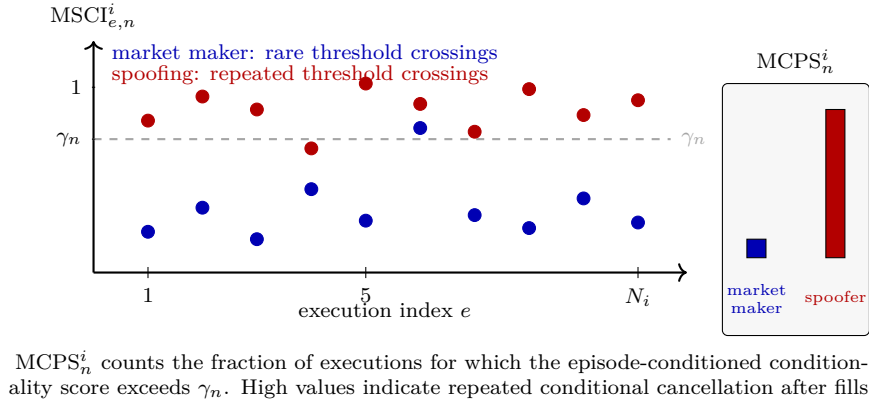


Figure 5: Operational interpretation of the Multilevel Conditional Probability Score. Each point is one execution for the agent. The score increases when the multilevel conditionality index repeatedly exceeds the threshold γ_n .

In practice, the depth n should be chosen according to the venue, tick size, and typical liquidity distribution of the instrument. A robust surveillance implementation can compute MCPS on a grid $\mathcal{N} = \{1, 2, 3, 5, 10\}$ and retain the resulting vector

$$(\text{MCPS}_1^i, \text{MCPS}_2^i, \text{MCPS}_3^i, \text{MCPS}_5^i, \text{MCPS}_{10}^i) \quad (23)$$

as an interpretable multiscale alert profile. A spoofer operating close to the touch should be detected at small n , whereas a strategy that layers deceptive volume deeper in the book should produce larger values only once deeper levels are included.

5.1.5 Price-Response Diagnostics

The conditionality scores above identify whether a client repeatedly removes same-client opposite-side liquidity after own executions. They do not, by themselves, show that the liquidity layer was economically useful. A stronger spoofing-compatible episode should also be associated with a short-horizon price response in the direction that improves the bona fide execution, followed by partial reversion once the suspicious layer is removed. We therefore complement MSCI, WM-

SCI, and MCPS with price-response diagnostics. These diagnostics are not a causal estimate of manipulation; they are economic consistency checks that test whether the observed conditional withdrawal coincides with the price movement that the strategy is meant to induce.

Let $M(t)$ denote a market price proxy, such as the mid-price or the standard queue micro-price, and let t_0 be the first observed posting time of the candidate deceptive layer used for event e . Define the execution direction

$$h_e = \begin{cases} +1, & \text{if the bona fide execution is a sell execution,} \\ -1, & \text{if the bona fide execution is a buy execution.} \end{cases} \quad (24)$$

With this convention, positive signed price changes are favorable to the executed order: a sell benefits from an upward move, whereas a buy benefits from a downward move. The pre-fill favorable movement is

$$\text{FPM}_e^i = h_e [M(t_e^-) - M(t_0)], \quad (25)$$

and the post-cancellation reversion is

$$\text{REV}_e^i = h_e [M(t_e^-) - M(t_e^+)]. \quad (26)$$

Thus $\text{FPM}_e^i > 0$ means that the market moved in the direction favorable to the small execution while the candidate layer was present, and $\text{REV}_e^i > 0$ means that this favorable movement partially reversed after the cancellation window. Finally, for the execution price P_e , define a local execution advantage relative to the pre-posture benchmark:

$$\text{ADV}_e^i = h_e [P_e - M(t_0)]. \quad (27)$$

This quantity is positive when a sell executes above the benchmark or a buy executes below the benchmark.

In the empirical implementation we compute these diagnostics using both the mid-price and the queue microprice. They are reported separately from MSCI and WMSCI rather than multiplied into a single opaque detector. The reason is evidentiary: MCPS measures the persistence of conditional opposite-side withdrawal, whereas FPM, REV, and ADV measure whether the same episodes are accompanied by the price response that would make the strategy profitable. A client-session with high MCPS but no favorable price response may still be a surveillance concern, but it is weaker evidence of economically effective spoofing than a session with repeated conditional withdrawals, favorable pre-fill price movement, post-cancel reversion, and positive execution advantage.

5.1.6 Event Reconstruction

The above quantities should be computed from order-level data rather than from aggregate snapshots whenever possible. The reason is that a resting order may move from one level to another because the best quote changes, even if the agent has not cancelled or modified the order. For intent isolation, the relevant event is the removal of agent-owned liquidity after the execution of the bona fide order. The operational procedure is therefore:

1. reconstruct the active orders of agent i immediately before t_e ;
2. assign each active order to its side s and level $k \leq n$, retaining in the episode-conditioned profile only orders whose first observed posting time lies in $[t_e - \Theta, t_e]$;
3. compute $L_{e,n,\Theta}^{i,b}(t_e^-)$, $L_{e,n,\Theta}^{i,a}(t_e^-)$, and $DWI_{e,n,\Theta}^i(t_e^-)$;
4. update the book through the interval $[t_e, t_e + \Delta\tau]$ using executions, cancellations, and amendments;
5. compute the corresponding episode-conditioned post-event quantities at t_e^+ ;
6. evaluate $MSCI_{e,n}^i$ and $WMSCI_{e,n}^i$;
7. compute the signed price-response diagnostics between the first candidate-layer posting time, the execution, and the post-cancellation window;
8. aggregate repeated behavior through the corresponding client-session scores.

This order-level reconstruction prevents ordinary repricing of the book from being mechanically interpreted as cancellation of deceptive liquidity, thereby reducing false positives for legitimate market makers.

As a diagnostic case, the all-sample run below uses a short 10-second post-execution cancellation window and a 90-second candidate-order age filter. Under this stricter timing rule, many matched events still have small or muted MSCI values because the distance-weighted profile does not always produce a large net DWI collapse, while WMSCI remains positive when a small execution is followed by a large same-client opposite-side withdrawal. This motivates reporting WMSCI alongside MSCI rather than treating either score as a complete detector in isolation.

6 Results

6.1 Cross-Instrument Detection Summary

[Table 3](#) reports the exploratory top-5 results obtained by applying the same episode-conditioned reconstruction to the three samples with the full-sample empirical depth kernels from [table 2](#) at $H = 10$ seconds. A matched event is a small passive execution followed by same-client cancellation of an opposite-side candidate order inside the 10-second post-execution window. The price response columns report the fraction of matched events with positive signed mid-price favorable pre-fill movement ($FPM > 0$) or post-cancel reversion ($REV > 0$). These quantities are descriptive surveillance diagnostics, not labels of manipulation.

The first result is that matched-withdrawal evidence is not confined to one instrument. The same rule identifies 219 matched events in Risanamento, 573 in Nexi, and 364 in Ferrari, involving 10, 9, and 5 distinct clients, respectively. The absolute number of matched events is largest for Nexi, whereas the most extreme event severity is observed in Risanamento, where the maximum WMSCI reaches 93.09 and the largest withdrawal-to-fill ratio in the matched set reaches 18,200. Ferrari produces fewer clients and less extreme WMSCI values, but still shows repeated matched withdrawals under the same reconstruction rule.

Table 3: Exploratory top-5 spoofing-compatible detections across the three samples. Matched events are small passive executions followed by same-client cancellation of an opposite-side candidate order inside the post-execution window. The price-response shares report the fraction of matched events with positive signed mid-price favorable pre-fill movement or post-cancel reversion.

Instrument	Executions	Matched events	Clients	Alerts	Max WMSCI	FPM > 0	REV > 0
Risanamento	15,341	219	10	4	93.09	44.7%	13.2%
Nexi	5,402	573	9	5	25.70	62.1%	30.7%
Ferrari	3,498	364	5	5	14.75	58.0%	33.5%

The second result is that WMSCI severity and price-response consistency are not the same signal. Risanamento contains the largest WMSCI events and the most extreme withdrawal-to-fill ratios, but only 44.7% of matched events have positive favorable pre-fill mid-price movement and only 13.2% have positive post-cancel mid-price reversion. Nexi and Ferrari have lower maximum WMSCI values, but materially stronger price-response consistency: approximately 62% and 58% of matched events have favorable pre-fill movement and roughly 31% and 34%, respectively, have post-cancel reversion. This supports the decision to keep matched-withdrawal severity, repetition, and price response as separate evidentiary dimensions instead of collapsing them into a single proof-of-intent score.

6.2 Client-Level Concentration

The matched events are concentrated in a small number of client identifiers. Table 4 reports the largest client-level clusters by instrument. The purpose of this table is not to identify a manipulator, but to show which client sessions an analyst would inspect first and which component of the evidence is driving prioritization.

Table 4: Largest client-level matched-withdrawal clusters by instrument. W/F is the maximum withdrawal-to-fill ratio among the client’s matched events. FPM > 0 and REV > 0 are client-level fractions of matched events with positive signed mid-price favorable movement and reversion.

Instrument	Client	Events	Max WMSCI	Mean WMSCI	Max W/F	FPM > 0 / REV > 0
Risanamento	17295	130	93.09	4.80	18,200	35.4% / 14.6%
Risanamento	0	43	6.10	1.34	31.8	79.1% / 14.0%
Risanamento	1000004133	37	13.94	2.00	88.5	40.5% / 10.8%
Nexi	23523	322	25.70	2.18	2,100	68.3% / 19.6%
Nexi	0	160	10.70	1.55	96.2	64.4% / 50.6%
Nexi	31425	62	23.50	4.14	140.7	16.1% / 22.6%
Ferrari	23523	214	14.75	1.23	53	59.3% / 19.6%
Ferrari	0	132	13.20	0.99	48	59.1% / 56.8%
Ferrari	10250181	10	2.86	1.25	6.2	30.0% / 0.0%

The Risanamento evidence is dominated by client 17295, which accounts for 130 matched events and contains the largest WMSCI event in the sample. However, its price-response diagnostics are mixed: only 35.4% of its matched events show a positive favorable pre-fill mid-price movement and 14.6% show positive reversion. Client 0 shows a higher favorable pre-fill movement share, while client 1000004133 has fewer matched events but larger WMSCI severity than

client 0. These patterns are strong matched-withdrawal cues, but they do not by themselves establish economically effective price pressure.

Nexi and Ferrari show a different profile. In both instruments, client 23523 is the largest matched-event cluster, while the anonymized or missing client identifier “0” also appears frequently. The repeated appearance of client 23523 across Nexi and Ferrari is a useful surveillance cue because it suggests that the pattern is not isolated to a single instrument. At the same time, the price-response shares differ by client: in Nexi, client 23523 has many matched events and high WMSCI severity, whereas client 0 has a higher fraction of post-cancel reversions. This illustrates why the event review dashboard should preserve order-level context, LOB snapshots, and the full zoom-window event table rather than ranking clients only by a scalar score.

6.3 Analyst Alert Prioritization

The final “alert” column in [table 3](#) is an analyst-prioritization output, not a calibrated probability of manipulation. In the prototype implementation, client sessions are first filtered by a minimum number of repeated matched-withdrawal events. The remaining sessions are ranked by a simple heuristic alert score that uses only the primary review evidence: repeated matched withdrawal, maximum event severity, and price-response consistency:

$$A_i = 0.50 \text{MES}_i + 0.30 \frac{\min\{\log(1 + \max_e \text{WMSCI}_e^i), 5\}}{5} + 0.20 \frac{\text{FPM}_i^+ + \text{REV}_i^+}{2}, \quad (28)$$

where MES_i is the matched-event share, FPM_i^+ and REV_i^+ are the fractions of matched events with positive signed mid-price favorable movement and reversion. MSCI and side-symmetry diagnostics remain available as secondary context, but they do not create a primary matched-withdrawal alert without repeated WMSCI evidence. The weights in [equation \(28\)](#) are deliberately heuristic; they only rank cases for human review. They should be recalibrated or replaced when analyst labels, negative controls, or venue-specific supervision objectives become available.

With the current thresholds, the prototype produces 4 analyst-review alerts in Risanamento, 5 in Nexi, and 5 in Ferrari. These counts should be interpreted as workload estimates under the chosen parameterization. They are useful for triage design because they connect detector sensitivity to a finite number of cases that can be inspected with the event-review dashboard, but they should not be interpreted as the number of spoofing episodes in the data.

References

- [1] J. Chen, P.-F. Hsieh, and J. J. Yang. Order spoofing, price impact, and market quality. *Pacific-Basin Finance Journal*, 97:103077, 2026.
- [2] P. Debie, C. Gardebroek, S. Hageboeck, P. van Leeuwen, L. Moneta, A. Naumann, J. M. E. Pennings, A. A. Trujillo-Barrera, and M. E. Verhulst. Unravelling the JPMorgan spoofing case using particle physics visualization methods. *European Financial Management*, 29(1):288–326, 2023.
- [3] European Parliament and Council of the European Union. Regulation (eu) no 596/2014

of the european parliament and of the council of 16 april 2014 on market abuse (market abuse regulation). <https://eur-lex.europa.eu/eli/reg/2014/596/oj/eng>, 2014. Official Journal of the European Union, L 173, 12 June 2014, p. 1.

- [4] T. Fabre and D. Challet. Learning the spoofability of limit order books with interpretable probabilistic neural networks. arXiv preprint arXiv:2504.15908, 2025.
- [5] L. Kang, T.-J. Mu, and X. Ning. Conspiracy spoofing orders detection with transformer-based deep graph learning. In Advanced Data Mining and Applications, Lecture Notes in Computer Science, pages 489–503, 2023.
- [6] L. Kang, T.-J. Mu, and X. Ning. Spoofing transaction detection with group perceptual enhanced graph neural network. In Machine Learning and Knowledge Discovery in Databases. Applied Data Science Track, Lecture Notes in Computer Science, pages 106–122, 2024.
- [7] E. J. Lee, K. S. Eom, and K. S. Park. Microstructure-based manipulation: Strategic behavior and performance of spoofing traders. Journal of Financial Markets, 16(2):227–252, 2013.
- [8] H. Li, M. Polukarova, and C. Ventre. Detecting financial market manipulation with statistical physics tools. In Proceedings of the 4th ACM International Conference on AI in Finance, pages 565–573, 2023.
- [9] L. Mendonça and A. De Genaro. Detection and analysis of occurrences of spoofing in the Brazilian capital market. Journal of Financial Regulation and Compliance, 28(3):369–408, 2020.
- [10] A. Stenfors, M. Doraghi, C. Soviany, M. Susai, and K. Vakili. Cross-market spoofing. Journal of International Financial Markets, Institutions and Money, 83:101735, 2023.
- [11] A. Stenfors and M. Susai. Spoofing and pinging in foreign exchange markets. Journal of International Financial Markets, Institutions and Money, 70:101278, 2021.
- [12] X. Tao, A. Day, L. Ling, and S. Drapeau. On detecting spoofing strategies in high-frequency trading. Quantitative Finance, 22(8):1405–1425, 2022.
- [13] J.-N. Tuccella, P. Nadler, and O. Şerban. Protecting retail investors from order book spoofing using a GRU-based detection model. arXiv preprint arXiv:2110.03687, 2021.
- [14] S. Xiang, Y. Jiang, Y. Chen, D. Cheng, G. Zhao, and C. Jiang. Generative dynamic graph representation learning for conspiracy spoofing detection. In Proceedings of the ACM Web Conference 2025, 2025.